

Hospital Indoor Navigation — POC Documentation

Document Type: Management Presentation
Version: 1.0
Status: Phase 1 — Active POC
Prepared for: Hospital Leadership & IT Management

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1. Executive Summary

Hospitals are large, complex environments. Patients and visitors often struggle to find the right ward, OPD room, or department — leading to frustration, delays, and a poor experience.

This document outlines a **two-phase Indoor Navigation solution** designed to solve this problem in a cost-effective, scalable manner.

What We Are Building

Phase	Approach	Status
Phase 1	QR-based Smart Navigation	✅ Active POC
Phase 2	BLE Beacon Auto-Tracking	🔵 Future Enhancement

Key Highlights

- **Phase 1 (Current POC)** uses QR codes placed at key hospital checkpoints. Patients scan a QR code, and the system instantly shows them turn-by-turn directions to their destination — no expensive hardware required.
- **Phase 2 (Future)** upgrades to Bluetooth Low Energy (BLE) beacons for fully automatic indoor tracking, similar to how GPS works outdoors.
- **Goal:** Improve the patient navigation experience, reduce staff interruptions for directions, and lay the foundation for a smart hospital infrastructure.

- **Immediate Priority:** Validate the QR-based approach with management and staff before investing in hardware.

2. Phase 1 — QR Smart Indoor Navigation (Current POC)

Overview

Phase 1 is a **web-based navigation system** that guides patients from the hospital entrance to their destination using QR codes placed at key checkpoints (reception, lift lobbies, corridor junctions).

No app download is needed. Patients simply scan a QR code using their phone's camera.

How It Works — Step by Step

```
Step 1 → Patient arrives at hospital
Step 2 → Scans QR code at reception / entrance
Step 3 → Enters destination (e.g., "OPD Room 204")
Step 4 → System generates the shortest route
Step 5 → Live AR-style guide appears on screen
Step 6 → Patient follows animated arrows through corridors
Step 7 → At each checkpoint, patient scans next QR to re-confirm position
Step 8 → Patient arrives at destination ✓
```

Key Features

- **QR Checkpoints** — Placed at reception, lift lobbies, floor junctions, and key corridors
- **Konva Floor Maps** — Interactive, zoomable hospital floor maps rendered digitally
- **Live AR-Style Guide** — Animated directional arrows guide the user in real time
- **Lift / Floor Transitions** — Navigation handles multi-floor journeys (e.g., "Take Lift to Floor 3")
- **Turn-by-Turn Directions** — Clear instructions at every junction (e.g., "Turn left → walk 30m → enter Room 204")

User Experience Flow

```
[ Patient scans QR at Reception ]
    ↓
[ System detects location checkpoint ]
    ↓
[ Route is calculated on floor map ]
    ↓
[ Live AR-style arrows appear on screen ]
    ↓
[ Patient follows corridor-by-corridor guide ]
    ↓
[ Lift transition handled automatically ]
    ↓
[ Patient arrives at destination ] ✓
```

3. Phase 1 — Technology Stack

Layer	Technology	Purpose
Frontend Framework	React + Vite	Fast, lightweight web app
Floor Map Rendering	react-konva	Interactive hospital floor maps
QR Code Scanning	html5-qrcode	In-browser QR scanning (no app needed)
Backend Server	Node.js + Express.js	Route calculation and API handling
Map Data	JSON-based floor maps	Lightweight, editable map definitions
Route Engine	Custom pathfinding logic	Shortest path between checkpoints
Navigation UI	AR-style animated arrows	Corridor simulation and direction display

Design Principle: This stack was chosen to be lean, fast to deploy, and easy to maintain — ideal for a POC that needs to demonstrate value quickly without heavy infrastructure.

4. Phase 1 — Benefits

Why QR Smart Navigation is the Right Starting Point

- **Lowest Possible Cost** — No hardware purchases. QR codes cost pennies to print.
- **Fast Deployment** — Can be live within weeks, not months.
- **No Hardware Installation** — No drilling, no wiring, no facilities team dependency.
- **Works on Any Smartphone** — Patients use their own devices; no special equipment needed.
- **Runs in a Web Browser** — No app installation required for patients.
- **Easy Management Demo** — Simple, visual, and instantly understandable for stakeholders.
- **Good Patient Guidance** — Meaningful improvement over printed maps or asking staff.
- **Clear Upgrade Path** — The same backend infrastructure powers Phase 2 (BLE) with minimal rework.

5. Phase 1 — Limitations

Understanding the limitations helps set the right expectations for this POC phase.

- **Manual Checkpoint Scanning** — Patients must actively scan QR codes at each checkpoint; the system does not auto-detect their location.
- **Not Fully Automatic** — Unlike GPS, there is no continuous real-time tracking between scans.
- **Position Updates Only at Scan Points** — If a patient takes a wrong turn between checkpoints, the system cannot self-correct until the next scan.
- **Limited Analytics** — Basic usage tracking only; no detailed patient flow data.
- **Depends on Patient Action** — Navigation quality depends on patients remembering to scan at each checkpoint.

These limitations are fully expected at POC stage and are addressed in Phase 2 (BLE).

6. Phase 1 — Cost Estimate

Budget Breakdown (INR)

Item	Estimated Cost (INR)
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QR Code Printing (laminated, weatherproof)	₹2,000 – ₹5,000
QR Code Standees / Wall Mounting	₹3,000 – ₹8,000
Development (POC build)	₹40,000 – ₹80,000
Cloud Hosting (first 6 months)	₹5,000 – ₹12,000
Hospital Floor Map Digitisation (sample)	₹8,000 – ₹15,000
Testing & QA	₹5,000 – ₹10,000
Approximate Total	₹63,000 – ₹1,30,000

Note: This is an approximate estimate for a single-floor or two-floor POC covering one wing or OPD block. Full hospital rollout costs will be scoped separately after POC validation.

Verdict

✔ **This is the best low-cost POC option available.** For under ₹1.5 lakh, the hospital gets a working, demonstrable patient navigation system.

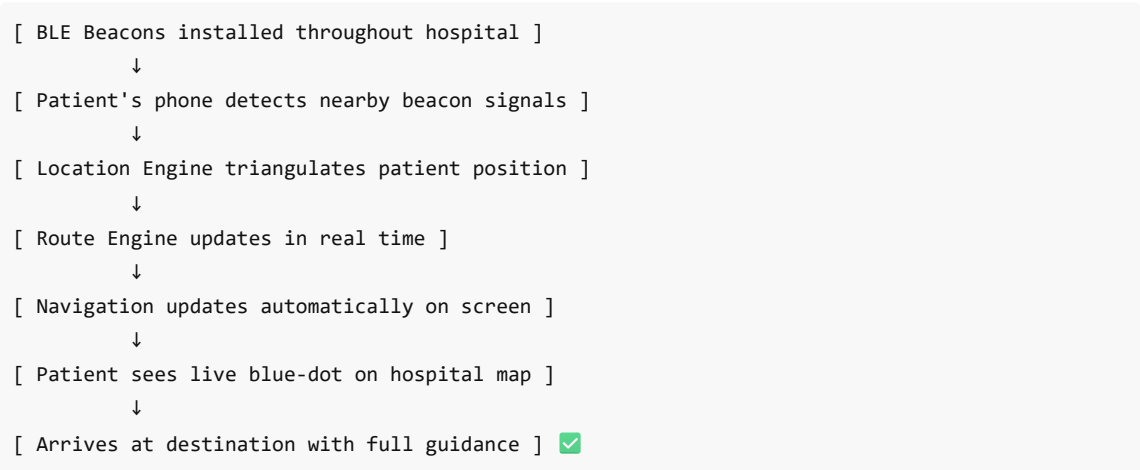
7. Phase 2 — BLE Beacon Indoor Navigation

What is BLE?

BLE (Bluetooth Low Energy) beacons are small, battery-powered wireless devices installed on walls and ceilings throughout the hospital. Each beacon continuously broadcasts a unique signal.

When a patient walks through the hospital with their phone, the phone automatically detects nearby beacons and calculates the patient's location — no scanning required.

How It Works



QR vs BLE — The Simple Difference

	QR Navigation	BLE Navigation
How location is detected	Patient scans QR at checkpoints	Phone detects beacons automatically

Tracking style	Manual, step-by-step	Continuous, automatic
Feel	Guided checkpoint navigation	Indoor GPS

Think of it this way: QR navigation is like following signposts. BLE navigation is like Google Maps — your position updates live as you move.

8. Phase 2 — BLE Technology Stack

Layer	Technology	Purpose
BLE Beacons	iBeacon / Eddystone hardware	Physical location broadcast devices
Mobile App	React Native (iOS + Android)	Native app for accurate beacon scanning
Location Engine	Beacon triangulation algorithm	Calculates real-time patient position
Floor Map Rendering	react-konva / MapLibre	Interactive multi-floor maps
Backend	Node.js + PostgreSQL	Location data, routes, analytics storage
Analytics Dashboard	Custom web dashboard	Patient flow, heatmaps, usage statistics
Future AR Layer	ARCore (Android) / ARKit (iOS)	Augmented reality corridor overlay

9. Phase 2 — BLE Benefits

BLE unlocks a significantly richer experience for patients and operational intelligence for the hospital.

Patient Experience

- **Automatic Location Updates** — No scanning needed; navigation updates as the patient walks
- **Live Blue-Dot Tracking** — Patient sees their live position on the hospital map
- **Better Accuracy** — Position updates every few seconds, not just at checkpoints
- **Seamless Multi-Floor Navigation** — Detects floor changes automatically via elevator beacons

Hospital Operations

- **Patient Flow Analytics** — Understand how patients move through the hospital
- **Congestion Heatmaps** — Identify bottlenecks in corridors or waiting areas
- **Staff Optimisation** — Deploy staff where patient volume is highest
- **Asset Tracking (Future)** — Track wheelchairs, equipment, and medical carts
- **Emergency Routing** — Real-time rerouting during emergencies or restricted areas

10. Phase 2 — BLE Limitations

Limitation	Details
Hardware Cost	Each beacon costs ₹800–₹2,500; large hospitals need 100–500 units
Installation Effort	Beacons must be physically mounted and configured throughout the hospital
Calibration Required	Signal mapping (radio frequency calibration) needed per floor

Native App Preferred	Web browsers have limited Bluetooth access; a mobile app is recommended
Signal Interference	Metal structures, thick walls, and medical equipment can affect accuracy
Ongoing Maintenance	Battery replacement and signal recalibration needed periodically
Higher Project Timeline	Full deployment takes 2–4 months vs. weeks for QR

11. Phase 2 — BLE Budget Estimate

Small Hospital / Single Building

Item	Estimated Cost (INR)
BLE Beacons (30–50 units @ ₹1,500 avg.)	₹45,000 – ₹75,000
Hardware Installation & Mounting	₹15,000 – ₹30,000
Signal Calibration & Radio Mapping	₹20,000 – ₹40,000
Mobile App Development (iOS + Android)	₹1,50,000 – ₹3,00,000
Backend + Analytics Integration	₹80,000 – ₹1,50,000
Total Estimate	₹3,10,000 – ₹5,95,000

Medium Hospital / Multi-Wing

Item	Estimated Cost (INR)
BLE Beacons (100–250 units)	₹1,50,000 – ₹3,75,000
Full Installation + Calibration	₹80,000 – ₹1,50,000
App + Backend + Analytics	₹3,00,000 – ₹5,00,000
Total Estimate	₹5,30,000 – ₹10,25,000

Enterprise / Large Hospital

Item	Estimated Cost (INR)
250–500 Beacons (or UWB/RTLS system)	₹5,00,000 – ₹12,00,000
Large-Scale Installation	₹2,00,000 – ₹5,00,000
Advanced Analytics + Integration	₹5,00,000 – ₹15,00,000
Total Estimate	₹12,00,000 – ₹32,00,000+

12. QR vs BLE — Comparison

Criteria	Phase 1 — QR Navigation	Phase 2 — BLE Navigation
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Cost	✔ Very Low (< ₹1.5L POC)	● Medium–High (₹3L–₹32L+)
Deployment Speed	✔ 2–4 weeks	⚠ 2–4 months
Hardware Required	✔ None (only printed QR)	⚠ Beacons required
Tracking Type	Checkpoint-based (manual scan)	Continuous auto-tracking
User Action Needed	Scan at each checkpoint	None — fully automatic
Patient Experience	Good	Excellent
Analytics	Basic	Advanced (heatmaps, flow)
Works in Web Browser	✔ Yes	⚠ Limited (app preferred)
Multi-Floor Support	✔ Yes (manual)	✔ Yes (automatic)
Maintenance	✔ Very Low	⚠ Moderate (batteries, calibration)
Scalability	Medium	High
Ideal For	POC, early adoption, budget-conscious	Full deployment, premium experience

13. Recommended Rollout Plan

Four-Stage Strategy

STAGE 1	STAGE 2	STAGE 3	STAGE 4
Weeks 1–4	Month 2–3	Month 4–6	Month 7+
QR POC (Internal)	Real QR Deployment	BLE Pilot (One Wing)	Full Hospital

Stage Details

Stage 1 — QR POC (Weeks 1–4)

- Deploy QR navigation system in OPD block or one wing
- Internal demo for management and key staff
- Collect feedback from early users
- Validate core navigation logic

Stage 2 — Real QR Deployment (Month 2–3)

- Expand QR system to full hospital (all floors, all departments)
- Print and install laminated QR codes across all checkpoints
- Launch to patients and measure adoption
- Gather navigation accuracy and satisfaction data

Stage 3 — BLE Pilot (Month 4–6)

- Install BLE beacons in one selected wing

- Run QR and BLE systems in parallel for comparison
- Measure user preference and accuracy improvements
- Calibrate beacon placement and signal mapping

Stage 4 — Full Analytics & AR Navigation (Month 7+)

- Full BLE rollout across entire hospital
- Activate advanced analytics dashboard
- Enable patient flow monitoring and heatmaps
- Explore AR overlay features for premium navigation experience

14. Final Recommendation

Our Recommendation

Start with QR Smart Navigation POC immediately. Add BLE only after management approval and demonstrated adoption success.

Why This Approach Makes Sense

The QR-based Phase 1 system is not just a shortcut — it is the strategically correct first step. Here is why:

1. Prove the concept first, invest later.

Before spending ₹5–30 lakhs on hardware and infrastructure, validate that patients will actually use the navigation system. QR does this for under ₹1.5 lakhs.

2. Fast time to value.

QR can be live in weeks. BLE takes months. Getting something real in front of management and patients quickly builds confidence and momentum.

3. The foundation carries forward.

The backend route engine, floor maps, and navigation logic built in Phase 1 are reused in Phase 2. Nothing is wasted.

4. Low risk.

If adoption is lower than expected, the hospital has spent very little. If adoption is strong, the case for BLE investment becomes self-evident.

5. Management alignment.

A working demo is worth more than any proposal document. Phase 1 gives management something tangible to see, test, and approve before committing to Phase 2 budgets.

Decision Summary

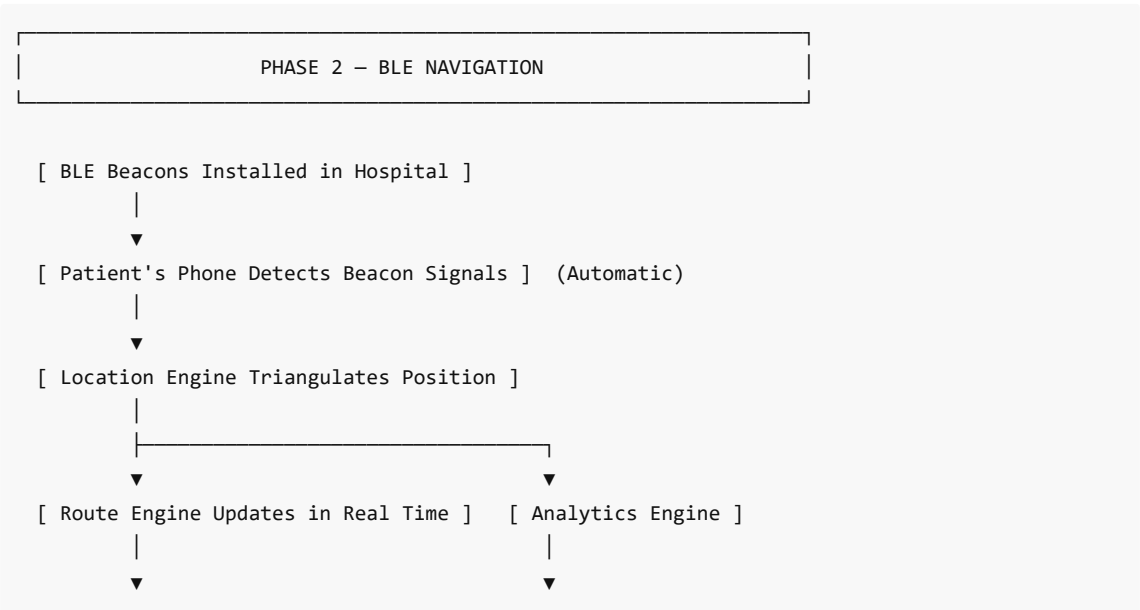
Decision Point	Recommendation
Start Phase 1 now?	✅ Yes — immediately
Full QR hospital rollout?	✅ Yes — after POC validation
Start Phase 2 (BLE)?	🟦 After Phase 1 adoption is confirmed
Budget for Phase 1	₹63,000 – ₹1,30,000
Budget for Phase 2	₹3,10,000 – ₹10,25,000 (scope-dependent)

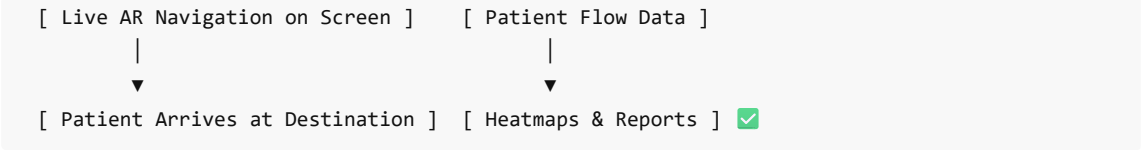
15. Visual Architecture Diagrams

Phase 1 — QR Navigation Architecture



Phase 2 — BLE Navigation Architecture





Combined System View

